www.tntextbooks.in

sequence of two statements S1 and S2 such that the input-output relation of S1, say R, is the input property of S2.

1. -- P
2. S1
3. -- R
4. S2
5. -- Q

Example 7.1. Let us solve the Farmer, Goat, Grass, and Wolf problem of Example 6.12. We decided to represent the state of the process by four variables farmer, goat, grass, and wolf, representing the sides of the farmer, goat, grass and wolf, respectively. In the initial state, all four variables have the value L (Left side). In the final state, all four variables should have the value R (Right side). The goal is to construct a statement S so as to move from the initial state to the final state.

1. **-- farmer, goat, grass, wolf = L, L, L, L**
2. **S**
3. **-- farmer , goat , grass , wolf = R, R, R, R**

We have to compose S as a sequence of assignment statements such that in none of the intermediate states

1. goat and wolf have the same value but farmer has the opposite value, or
2. goat and grass have the same value but farmer has the opposite value.Subject to these constraints, a sequence of assignments and the state after each assignment are shown in Figure 7.3.

1. -- farmer, goat, grass, wolf = L, L, L, L
2. **farmer, goat := R, R**
3. -- farmer , goat , grass , wolf = R, R, L, L
4. **farmer := L**
5. farmer, goat, grass, wolf = L, R, L, L
6. **farmer, grass := R, R**
7. -- farmer , goat , grass , wolf = R, R, R, L
8. **farmer, goat := L, L**
9. -- farmer, goat, grass, wolf = L, L, R, L
10. **farmer, wolf := R, R**
11. -- farmer , goat , grass , wolf = R, L, R, R
12. **farmer : = L**
13. -- farmer , goat , grass , wolf = L, L, R, R
14. **farmer , goat : = R, R**
15. -- farmer , goat , grass , wolf = R, R, R, R

Figure 7.3: Sequence of assignments for goat, grass and wolf problem